

Fileless Malware

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AGENDA

- What is Fileless Malware?
- Working of the Malware
- Example (Recent attacks)
- Detection and Prevention.
- Conclusion.

What is a Fileless Malware?

- Fileless malware is a type of malicious software that doesn't rely on traditional executable files to infect a system. Instead, it operates directly within a system's memory or uses legitimate system tools and processes to carry out its attacks. This makes it harder to detect since it doesn't leave behind traditional files on disk that can be easily scanned by antivirus or anti-malware tools.
- **Memory-based:**
- **No disk footprint:**
- **Leveraging legitimate tools**
- **Persistence methods:**

Attack methodologies

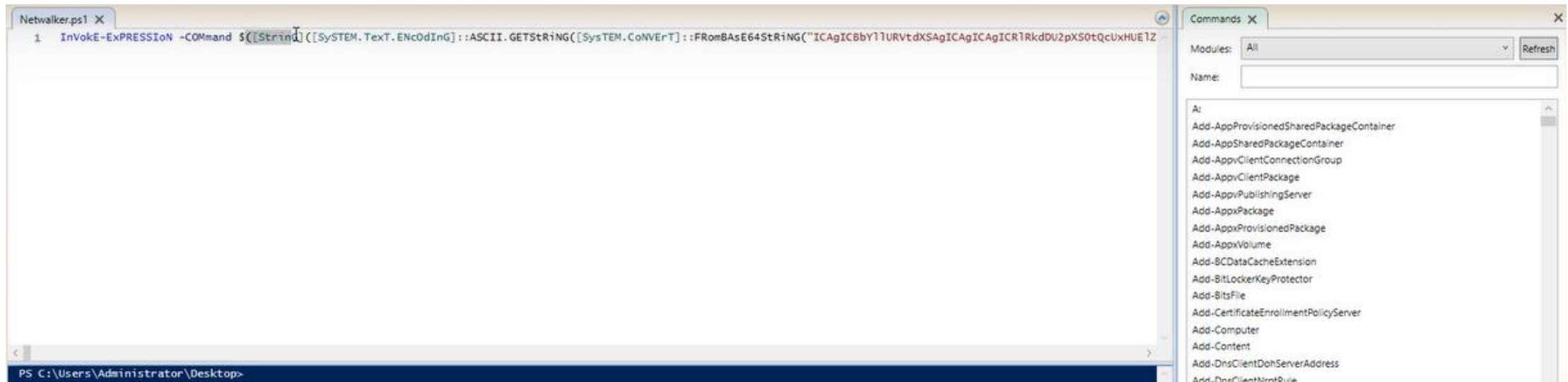
- Registry Manipulation
- Scheduled Tasks or Services
- WMI Persistence
- Exploitation of Software Vulnerabilities:
- Browser or Application Hijacking
- Living Off the Land (LotL):

Stages of a fileless attack

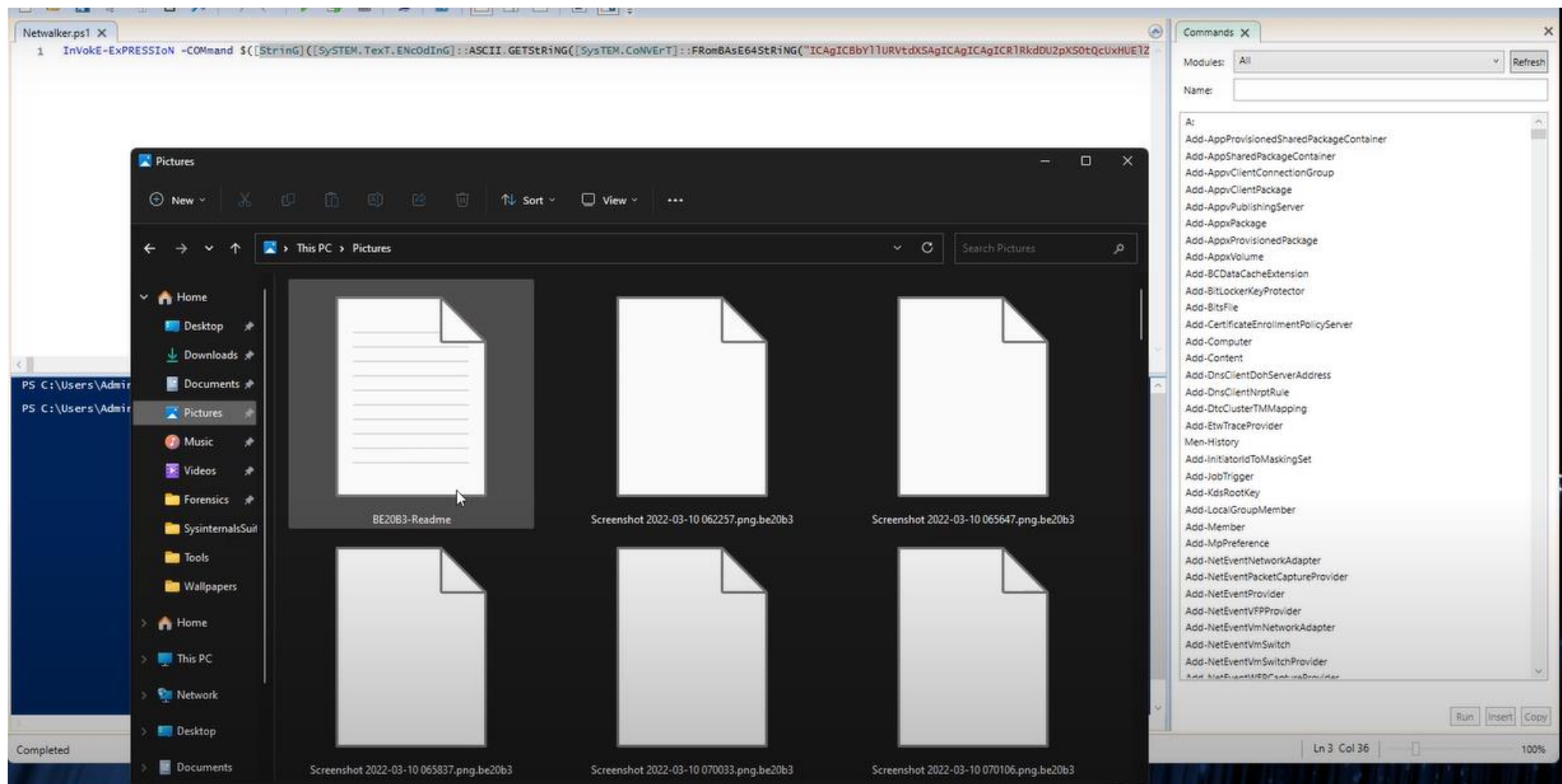
- Stage #1: Gain Access
- Technique: Remotely Exploit a vulnerability and use web scripting for remote access
- Stage #2: Steal Credentials
- Technique: Remotely Exploit a vulnerability and use web scripting for remote access
- Stage #3: Maintain Persistence
- Technique: Modify registry to create a backdoor
- Stage #4: Exfiltrate Data
- Technique: Uses file system and built-in compression utility to gather data , then uses FTP to upload the data

NETWALKER MALWARE





Encrypted powershell script



Files getting encrypted as the powershell command runs without the need of an executable


```
BE20B3-Readme.txt - Notepad
File Edit Format View Help
Hi!
Your files are encrypted by Netwalker.
All encrypted files for this computer has extension: .be20b3

--
If for some reason you read this text before the encryption ended,
this can be understood by the fact that the computer slows down,
and your heart rate has increased due to the ability to turn it off,
then we recommend that you move away from the computer and accept that you have been compromised.
Rebooting/shutdown will cause you to lose files without the possibility of recovery.

--
Our encryption algorithms are very strong and your files are very well protected,
the only way to get your files back is to cooperate with us and get the decrypter program.

Do not try to recover your files without a decrypter program, you may damage them and then they will be impossible to recover.

For us this is just business and to prove to you our seriousness, we will decrypt you one file for free.
Just open our website, upload the encrypted file and get the decrypted file for free.

---

Steps to get access on our website:

1.Download and install tor-browser: https://torproject.org/

2.Open our website: pb36hu4spl6cyjdfhing7h3pw6dhp32ifemawkujj4gp33ejzdq3did.onion
If the website is not available, open another one: rnfdsgm6wb6j6su5txkek4u4y47kp2eatvu7d6xhyn5cs4lt4pdrqqd.onion

3.Put your personal code in the input form:

{code_be20b3:
USR7srKOIOVQ+OjaQb6F2YxMB1ufmR1GwM+sJsAYcWbaXKnwt7
7KcLfby0uEJgZeBkUHgwWnFCHjow7tSioW4TQX3KTqjgx0L0Sb
SHrhjX93NzSHCY/tSULwPccj4WkuN+a/vDBmR4HwUnmIDDbz9t
ipCK8uuD2WPtiEwgnn0zAo56CM1Sc0sb/evN2AwoXU50AW0d7k
```

31
/ 60

Community Score

31 security vendors and 2 sandboxes flagged this file as malicious

f4656a9af30e98ed2103194f798fa00fd1686618e3e62fba6b15c9959135b7be

5.10 MB
Size

2022-09-06 16:37:47 UTC
2 months ago

Netwalker.ps1

long-base64

powershell

DETECTION

DETAILS

RELATIONS

BEHAVIOR

CONTENT

TELEMETRY

COMMUNITY 30 +

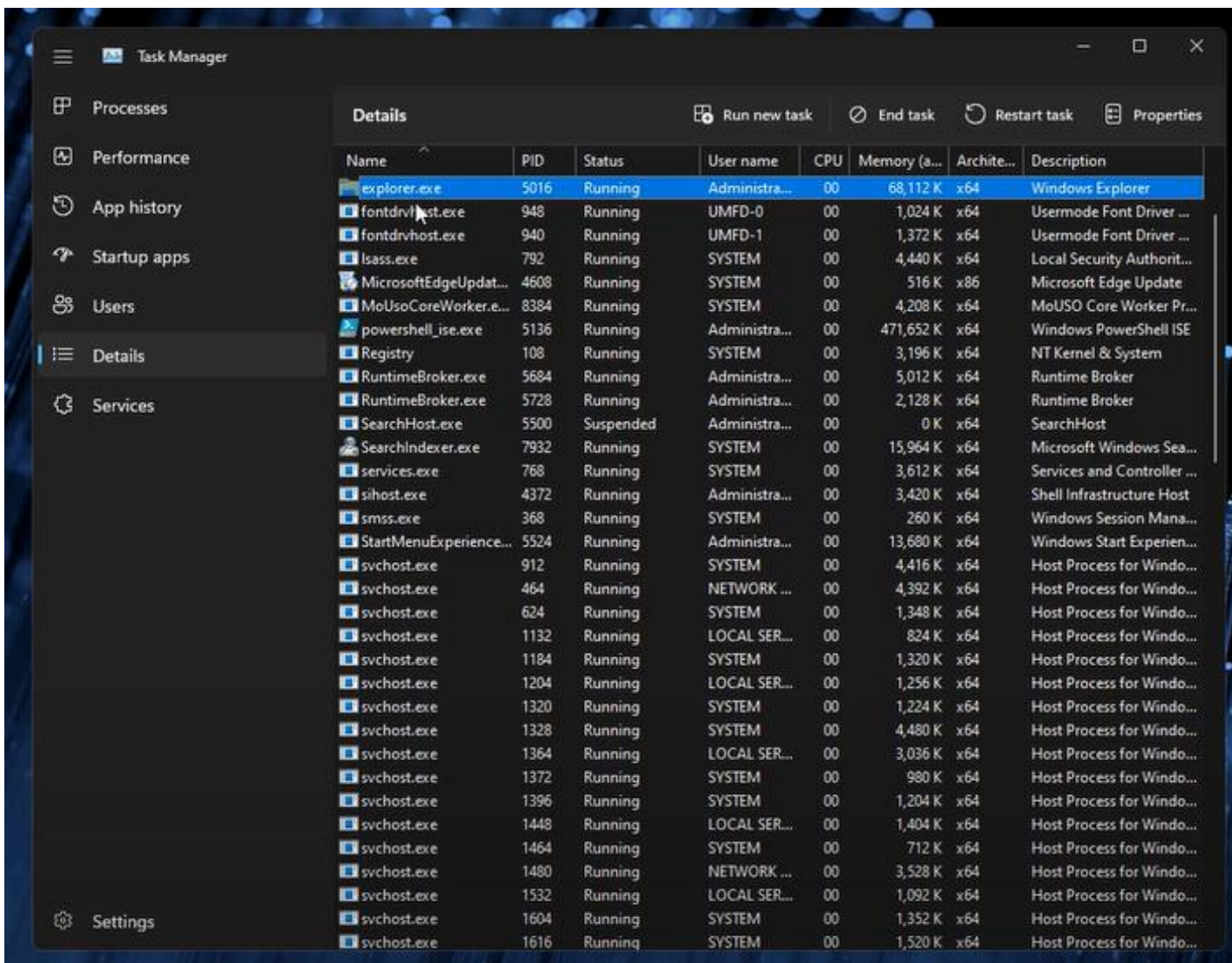
Dynamic Analysis Sandbox Detections

The sandbox Dr.Web vxCube flags this file as: RANSOM , MALWARE

The sandbox DAS-Security Orcas flags this file as: MALWARE

Security vendors' analysis on 2022-09-06T16:37:47 UTC

Ad-Aware	Trojan.Ransom.GenericKD.43121546	AhnLab-V3	Trojan/PS.Ransom
ALYac	Trojan.Ransom.Powershell	Antiy-AVL	Trojan/Generic.ASSuf.2C7A6
Avast	Other:Malware-gen [Trj]	AVG	Other:Malware-gen [Trj]
BitDefender	Trojan.Ransom.GenericKD.43121546	Comodo	Malware@#3plx81z3yiru
DrWeb	Trojan.Encoder.31757	Emsisoft	Trojan.Ransom.GenericKD.43121546 (B)
eScan	Trojan.Ransom.GenericKD.43121546	ESET-NOD32	Win64/Filecoder.Netwalker.A
Fortinet	BAT/Agent.61E2ltr	GData	Script.Trojan-Ransom.Netwalker.2Z0X.JW
Google	Detected	Ikarus	Trojan-Ransom.NetWalker
Kaspersky	HEUR:Trojan.PowerShell.Generic	Lionic	Trojan.PowerShell.Generic.4lc
MAX	Malware (ai Score=100)	McAfee	PS/Netwalker.b
McAfee-GW-Edition	PS/Netwalker.b	Microsoft	Ransom.PowerShell/NetWalker!MTB



Can only see explore.exe in task manager that basically is running the script



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Ransomware

Reflective Loading Runs Netwalker Fileless Ransomware

Ransomware in itself poses a formidable threat for organizations. As a fileless threat, the risk is increased as it can more effectively evade detection. We discuss how Netwalker ransomware is deployed filelessly through reflective DLL injection.

By: Karen Victor
May 18, 2020
Read time: 5 min (1290 words)



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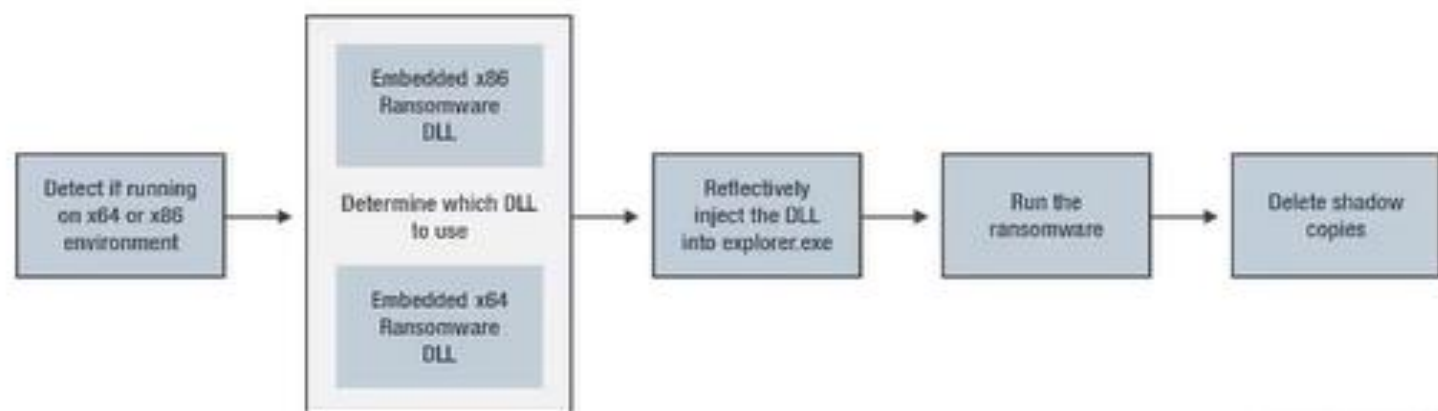
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Threat actors are continuously creating more sophisticated ways for malware to evade defenses. We have observed Netwalker ransomware attacks that involve malware that is not compiled, but written in PowerShell and executed directly in memory and without storing the actual ransomware binary into the disk. This makes this ransomware variant a **fileless threat**, enabling it to maintain persistence and evade detection by abusing tools that are already in the system to initiate attacks.

Related Articles

[Earth Preta Spear-Phishing Governments Worldwide](#)
[Pillfered Keys: Free App Infected by Malware Steals Keychain Data](#)

Analysis of the PowerShell Script



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Figure 1. Overview of the PowerShell script's behavior

Fileless Malware Detection

- IOAs focus on detecting malicious intent, not the specific attack method or steps.
- They analyze actions like code execution, lateral movement, and cloaking attempts.
- IOAs are effective against fileless malware that evades signature-based and machine learning detection.
- They monitor event sequences to uncover the true goals of an attack, regardless of the tools used.
- IOAs can detect malicious activities, even when attackers use legitimate accounts or stolen credentials.
- Managed threat hunting services offer 24/7 proactive monitoring, detecting subtle intrusions that standard security tools may miss, making them ideal for defending against fileless malware.
- Outsourcing threat hunting to expert providers saves organizations time and resources, as it involves extensive data gathering and analysis critical for identifying sophisticated attacks.

Key Points for A SOC analyst.

- Behavioral Analysis and Anomaly Detection
- Endpoint Detection and Response (EDR)
- Monitoring Indicators of Attack (IOAs)
- PowerShell and Script Logging
- Network Traffic Analysis
- Memory Forensics

Fileless Malware prevention

- Enable Advanced Script Logging
- Behavioral Monitoring and EDR Solutions
- Apply Least Privilege Principle
- Keep Software and Systems Updated
- Network Segmentation and Traffic Monitoring
- Disable Unnecessary Features
- Security Awareness Training

Conclusion

- Fileless malware is a sophisticated and stealthy form of cyberattack that operates directly in memory or by abusing legitimate system tools, making it difficult to detect using traditional signature-based methods. Its ability to avoid leaving a disk footprint poses a significant challenge to cybersecurity defenses. However, with a focus on behavior-based detection, proactive threat hunting, and preventive measures like script logging, endpoint monitoring, and limiting privileges, organizations can effectively detect and mitigate the risks posed by fileless malware.